

Bobby Flennoy

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EDUCATION

University of Texas at Arlington

Bachelor of Science in Computer Science, Minor in Cognitive Neuroscience

Aug. 2021 – Aug. 2026

Arlington, TX

TECHNICAL SKILLS

- **Languages:** JavaScript, TypeScript, Python, C++, HTML, CSS.
- **Frameworks & Libraries:** React, Next.js, Node.js, FastAPI, Electron, D3.js, TailwindCSS.
- **Systems & Cloud:** GCP (Pub/Sub, Cloud Run, Functions), AWS, Docker, CI/CD, Redis, Load Balancing, Load Testing.
- **Backend & Security:** Supabase (Postgres), JWT Security, OAuth2, REST APIs, Schema Design.
- **Core CS:** Data Structures & Algorithms (A*, BFS, DFS), Compiler Design, Git.

EXPERIENCE

Product Developer & Researcher

Pearl Discovery (Non-Profit)

Mar. 2026 – Present

Remote

- **Expert Review Tool:** Built the backend for an auditing tool where human experts verify AI labels, creating a high-quality dataset for model fine-tuning.
- **Cost Reduction:** Swapped out expensive Claude 3.5 calls for local SLMs (Small Language Models) to handle routine text tasks, cutting monthly API costs by 90%.

UX Research Intern

CBRE

Jun. 2025 – Aug. 2025

Richardson, TX

- **User Insights:** Used MaxDiff modeling to survey 500+ users and find exactly where facility management software was failing, cutting the time to build working prototypes in half.
- **Frontend Prototypes:** Built functional React prototypes to test complex UI patterns and state management before moving them into full-scale production.

PROJECTS

TeachTrack: University Analytics Platform

- **The Problem:** Apps like Canvas and Blackboard focus on general instructor ratings instead of whether students actually learned the material, leaving departments with no real data for accreditation.
- **Backend:** Built a system on **Supabase & Postgres** that maps raw student feedback to official university learning outcomes.
- **Scaling:** Preparing for high-traffic production by setting up **Redis** for database caching and **Load Balancers** to handle heavy traffic during end-of-semester evaluation cycles.
- **Security:** Set up a **JWT-based invite system** where professors send secure, temporary links to students, keeping all data siloed and FERPA-compliant.

SmartDash: IoT Data Pipeline

- **The Problem:** Facility managers usually have to wait for things to break. Large property owners need a way to see real-time health for thousands of devices at once to prevent failures.
- **Data Flow:** Built a **GCP pipeline** using **Pub/Sub and Cloud Functions** to catch and process 1Hz telemetry streams from simulated hardware.
- **Storage:** Used **Firestore** for a live-state dashboard and **BigQuery** to store all historical logs for predictive maintenance and trend analysis.

TaskFlow: Datacenter Logistics (HackUTD 1st Place)

- **The Problem:** Technicians in massive 400MW facilities waste time on manual work orders and navigating server clusters.
- **The Logic:** Wrote a routing engine using *A and BFS algorithms** to find the fastest path for technicians to handle hardware tickets across the floor.
- **The App:** Built a web app for task management and an **iOS AR app** that helps technicians find exactly which server needs work by pointing their camera at the rack.

EXTRACURRICULAR

President: Association for Computing Machinery (ACM)

1st Place: HackUTD (NorthMark Cloud & Compute Challenge)

Creative Director: HackUTA

Aug. 2023 – Present

Nov. 2025

Aug. 2023 – Dec. 2025